

# Electron Configuration

Answer key with worked solutions

## A. Understanding

1. p subshell.

**3 orbitals**, each holding 2 electrons: **6 electrons** in total.

2. Pauli and Hund.

**Pauli:** an orbital holds at most two electrons, and they must have opposite spins. **Hund:** electrons in the same subshell occupy separate orbitals with parallel spins before any orbital is doubly occupied.

## B. Atoms

3. Full configurations.

(a) C:  $1s^2 2s^2 2p^2$  (b) Mg:  $1s^2 2s^2 2p^6 3s^2$  (c) Cl:  $1s^2 2s^2 2p^6 3s^2 3p^5$

4. Noble-gas configurations.

(a) Ca: **[Ar]  $4s^2$**  (b) Fe: **[Ar]  $3d^6 4s^2$**  (c) Br: **[Ar]  $3d^{10} 4s^2 4p^5$**

5. Identify the elements.

(a) 7 electrons: **nitrogen** (b)  $10 + 7 = 17$ : **chlorine** (c)  $18 + 14 = 32$ : **germanium**

## C. Orbital diagrams

6. Nitrogen.

2s:  $\uparrow\downarrow$  2p:  $\uparrow$   $\uparrow$   $\uparrow$ . By Hund's rule the three 2p electrons occupy separate orbitals: **3 unpaired electrons**.

7. Unpaired electrons.

(a) S,  $[\text{Ne}] 3s^2 3p^4$ : 3p is  $\uparrow\downarrow$   $\uparrow$   $\uparrow$ , so **2** (b) Ne,  $1s^2 2s^2 2p^6$ : all subshells full, so **0**

## D. Ions

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### 8. Ions.

(a)  $\text{O}^{2-}$ :  $8 + 2 = 10$  electrons,  $1s^2 2s^2 2p^6 = [\text{Ne}]$  (b)  $\text{Mg}^{2+}$ :  $12 - 2 = 10$  electrons,  $[\text{Ne}]$  (c)  $\text{Fe}^{3+}$ : remove both 4s electrons, then one 3d:  **$[\text{Ar}] 3d^5$** .  $\text{O}^{2-}$  **and**  $\text{Mg}^{2+}$  have the same configuration (they are isoelectronic with neon).

## E. Challenge

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### 9. Chromium.

A half-filled d subshell (one electron in each of the five 3d orbitals) is especially stable, so one 4s electron moves into 3d. Unpaired electrons: five in 3d plus one in 4s = **6**.